# Joint UAV Trajectory, Relay Selection, and Bandwidth Allocation for Deadline-Aware Alert Delivery in Disaster Networks

**Group 5**

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**Abstract**

After a disaster, alerts raised in the field must reach the rescue center before their deadlines even though parts of the terrestrial network have failed. Existing unmanned aerial vehicle (UAV) relay designs optimize rates or the number of served devices and treat the ground network as an ideal sink, ignoring both per-alert deadlines and the limited backhaul of the surviving relays. This paper jointly optimizes the trajectory of a single UAV relay, the path of every alert through a capacity-limited ground network, and the bandwidth allocation, so as to maximize the fraction of alerts delivered on time under a probabilistic line-of-sight (PrLoS) channel. We prove that the problem is NP-hard even for a fixed trajectory and derive a linear programming (LP) upper bound that holds for every channel realization and every plan whose paths come from a precomputed candidate set. We then propose a block search over paths, bandwidth weights, and trajectories whose blocks are local searches around the current plan: paths are rerouted around the links that the plan congests, and trajectory segments are shifted toward saturated UAV links within the speed limit. Every neighbor is scored by simulating the plan on sampled channel realizations, and a variant additionally repairs the schedule of alerts at risk of missing their deadlines. On 30 evaluation scenarios synthesized from Topology Zoo and SNDlib, the block search delivers 0.559 of the alerts on time, against 0.494 for a UAV on a fixed trajectory and 0.413 without a UAV. Its advantage over the fixed trajectory holds on all ten topologies, with a Holm-corrected $p$-value of 0.0156, and grows when the backhaul is narrow. Ablations attribute the gain to selecting paths and trajectory jointly and to designing on sampled realizations rather than expected rates, whereas schedule repair changes the timely-delivery ratio by only 0.002 and a leximin fairness objective lowers it by 0.190. The block search also exceeds an adaptation of the method of Tran et al. by 0.028 on the default scenarios and by 0.079 under narrow backhaul, although neither difference survives Holm correction, and the ranking of all methods is unchanged in a case study with 63 targets extracted from RescueNet imagery.

**Keywords:** UAV relay; trajectory optimization; relay selection; bandwidth allocation; deadline-aware delivery; linear programming bound; local search; paired statistical testing; RescueNet.

## 1. Introduction

Disasters routinely sever or degrade parts of the terrestrial communication network, while rescue teams, sensors, and residents in the affected area keep raising alerts that must be forwarded to a coordination center. The value of such an alert depends on whether it arrives before a given deadline: a report of a collapsed building that arrives late may no longer help to save anyone. The appropriate performance measure is therefore not the average transmission rate but the fraction of alerts whose data are delivered in full before their deadlines, which we call the timely-delivery ratio.

UAV relays are a natural means of reconnecting isolated areas with the surviving part of the network. Unlike terrestrial base stations, UAVs can be deployed on demand and establish

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line-of-sight links with ground terminals more readily, since they are less exposed to blockage by buildings and terrain. Huang et al. [1] exploit these properties by jointly optimizing the resource scheduling, transmit power, and trajectory of a UAV relay under the PrLoS channel model, using block coordinate descent (BCD) combined with successive convex approximation (SCA) to maximize the minimum average rate. Works that count the devices served before a deadline, such as those of Tran et al. [2] and Samir et al. [3], come closer to the alert-delivery objective, but they assume a single ground sink that accepts all data forwarded by the UAV.

Transferring these designs to deadline-aware alert delivery in a damaged network is challenging for three reasons. First, the number of timely alerts is a count of binary events that changes discontinuously with the decisions, so rate-based convex surrogates do not track it; in our preliminary tests, bandwidth schedules and paths taken from an LP relaxation did not improve the delivered count once executed by a realistic scheduler. Second, relay selection interacts with the backhaul: a relay close to a source may have lost its backhaul or be congested, and the UAV position with the best channel is not necessarily the one that delivers the most alerts on time. Third, the PrLoS channel is random, and designing on expected rates overestimates the achievable rate by Jensen's inequality [4], so a plan that looks feasible on average may miss many deadlines in practice.

Based on the aforementioned discussion, this paper investigates the joint optimization of the trajectory of one UAV, the relay path of each alert, and the bandwidth allocation, with the objective of maximizing the timely-delivery ratio; time-respecting connectivity to the center serves as a secondary objective in lexicographic order. We retain the three-block structure of [1] but replace each convex subproblem by an evaluated local search that modifies one part of the current plan: an alert's path is rerouted through another entry node or around a congested backhaul edge, and a segment of the trajectory is shifted within the speed limit toward the UAV links that are short of capacity. Every candidate change of paths, bandwidth weights, or trajectory is scored by simulating the plan with the actual scheduler on a small set of sampled channel realizations, which forms a sample average approximation of the expected objective, and the total number of such evaluations is capped by a budget. Scoring on samples addresses the first and third challenges at once, since it measures the exact count of timely alerts and exposes the plan to channel outcomes worse than the mean, while searching over paths and trajectory jointly addresses the second. To measure how much room for improvement remains within the candidate paths, we derive an LP upper bound that is valid for every channel realization on a given trajectory.

Simulations on scenarios synthesized from the public datasets Topology Zoo [5], SNDlib [6], and RescueNet [7] show that the block search raises the timely-delivery ratio over a UAV on a fixed trajectory on every topology, by a larger margin when the backhaul is narrow. Ablations reveal that the gain comes from choosing paths and trajectory jointly and from designing on sampled realizations, that the local trajectory moves perform on par with a scan of predefined tours while spending far fewer evaluations per round, and that the search routes a substantial share of the alerts outside the precomputed candidate paths; once the scheduler shares bandwidth according to the backlog, explicit bandwidth weights and a dedicated risk-driven repair mechanism add no measurable gain. The block search also outperforms an adaptation of the method of Tran et al. [2], although the differences are not significant after Holm correction, and the ranking of all methods is preserved when the sources are placed at damaged buildings extracted from RescueNet.

The main contributions of this paper are summarized as follows:

- We formulate the joint trajectory, relay-path, and bandwidth problem with per-alert release times, deadlines, and sizes, a ground relay network with finite backhaul, the PrLoS channel,

and slot-level causality, and we prove that the problem is NP-hard even when the trajectory is fixed.
- We derive a per-realization LP upper bound, prove its validity, and quantify its tightness against a mixed-integer linear program (MILP) on reduced instances.
- We propose a sample-scored block search whose path and trajectory blocks are local searches around the current plan: paths are rerouted around the links that the current plan congests, and trajectory segments are shifted toward saturated UAV links within the speed limit. A risk-driven repair variant and a leximin variant isolate the effects of repair and of the objective.
- We evaluate all methods on two scenario families with topology-level paired tests, ablations, budget and sensitivity studies, a comparison with an adapted state-of-the-art method, and a case study based on real post-disaster imagery.

## 2. Related Work

We review four lines of work related to our problem: UAV relaying and data collection under time constraints, backhaul-aware placement and station selection, the solution methods used for joint trajectory and resource design, and scheduling with release dates and deadlines. Table 1 compares the most relevant works along the attributes that matter for deadline-aware alert delivery; its entries are taken from the abstracts of the cited works and from the full text of [1], and "Not stated" marks an attribute that is not mentioned.

Huang et al. [1] use a UAV as a buffered decode-and-forward relay for multiple user pairs under the PrLoS channel and maximize the minimum average rate through BCD combined with SCA; our work inherits their channel model and three-block structure. The works closest to timely alert delivery maximize the number of devices served. Tran et al. [2] require the data of each device to reach a ground gateway in time via the UAV's memory and optimize bandwidth, power, and trajectory through binary-variable relaxation and inner approximation. Samir et al. [3] assign a hard deadline to each device, obtain the global optimum with branch, reduce and bound for small instances, and resort to SCA for large ones, while Du et al. [15] count the devices served within a finite collection time. Other works place time in the constraints or the objective in different ways: a common collection deadline with an MILP model and a heuristic [8], an age-of-information threshold together with the selection of the ground station that receives the data [9], [10], bounded queueing delay in an online Lyapunov-based design [11], delay as the objective of UAV relaying or of priority-based collection [12]–[14], and message time-to-live in a delay-tolerant network with reinforcement-learning UAVs [16].

When a UAV acts as a base station or relay connecting users to the core network, the backhaul capacity limits the number of users that can be served. Kalantari et al. [17] place flying base stations in 3D according to the backhaul type; Selim and Kamal [18] employ tethered drones as post-disaster backhaul; Yu et al. [19] jointly optimize bandwidth and placement under a free-space-optics backhaul capacity constraint; and Queiros et al. [20] allocate bandwidth and position second-tier flying relays given the known trajectories of the first tier. None of these works imposes per-request time constraints.

BCD combined with SCA is the prevailing approach to joint trajectory and resource design [1], [10], [21]; Wu et al. [21] prove that it converges to at least a locally optimal solution of the max–min throughput problem. Works that count served requests must handle binary variables, either by relaxation [2] or by branch-and-bound [3]. Park and Lee [4] show that approximating the expected rate by the rate at the average channel overestimates the true rate because of Jensen's inequality, which motivates our choice to score designs on sampled realizations. Large neighborhood search with destroy and repair operators [22] is a precedent for our repair mechanism, and routing over

**Table 1.** Comparison of related works.

| Work | Objective | Station selection, backhaul | Time constraint | Channel, uncertainty | Resources | Method |
|---|---|---|---|---|---|---|
| [1] | Max–min average message rate | No | No | PrLoS | Power, slot scheduling | BCD, SCA |
| [2] | Max. devices served on time | Not stated | Yes, per device | Not stated | Bandwidth, power, UAV memory | Binary relaxation, inner approximation |
| [3] | Max. devices served before deadline | Not stated | Yes, per device | Not stated | Radio resources | BRB, SCA |
| [8] | Min. collection energy | Not stated | Yes, collection deadline | Not stated | Not stated | MILP, heuristic |
| [9] | Min. mission time | Not stated | Yes, AoI threshold | Not stated | Not stated | Decomposition, SCA, GA |
| [10] | Min. mission time | Yes, UAV–station link | Yes, per data item | Not stated | Not stated | BCD, SCA |
| [11] | Min. UAV energy | Yes, user association | Yes, bounded queueing delay | Random demand, mobility | Not stated | Lyapunov, online decomposition |
| [12] | Min. total information delay | Not stated | Delay objective | Not stated | Channel assignment | Alternating Newton, relax-and-round, SCA |
| [13] | UAV energy and priority-weighted delay | Not stated | Priority-weighted delay objective | Mobile sensor nodes | Not stated | Online RL heuristic |
| [14] | Lower delay, higher relay throughput | Not stated | Queueing congestion delay | Not stated | Not stated | Queue-threshold trajectory control |
| [15] | Max. devices served within collection time | Not stated | Yes, collection time | Not stated | Time, transmit power | Alternating, SCA, quadratic approx. |
| [16] | Higher DTN message delivery | Yes, opportunistic routing | Yes, message TTL | Not stated | Not stated | PPO reinforcement learning |
| [17] | Max. users or sum rate | Yes, backhaul | Not stated | Not stated | Not stated | 3D placement search |
| [18] | Min. drone energy | Yes, tethered drone backhaul | Not stated | Not stated | Not stated | Placement and energy optimization |
| [19] | Bandwidth and flying-station placement | Yes, FSO backhaul | Not stated | Not stated | Bandwidth | BROAD algorithm |
| [20] | Network performance with minimum rate | Yes, flying backhaul relays | Not stated | Not stated | Bandwidth | Penalized simulated annealing |
| This work | Timely-delivery ratio | Yes, ground relays with finite backhaul | Yes, per-alert deadline | PrLoS, design on sampled realizations | Weighted bandwidth | Evaluated block search, schedule repair |

time-varying graphs with known dynamics [23] is a precedent for our time-respecting connectivity measure.

On a single link, selecting the set of alerts delivered on time resembles single-machine scheduling to minimize the number of late jobs. The non-preemptive version $1 \,|\, r_j \,|\, \sum U_j$ is strongly NP-hard [24], whereas the preemptive version $1 \,|\, r_j, \mathrm{pmtn} \,|\, \sum U_j$ is solvable in polynomial time by dynamic programming [25]. Our model allows data to be split across slots, which is closer to the preemptive case; as we show, its hardness stems instead from the requirement that each alert follow exactly one path. Our upper bound relies on the concavity of the rate in the bandwidth and on its envelope of tangents, a standard tool of convex optimization [26].

As Table 1 shows, no existing work combines per-request deadlines with the selection of ground relays over a backhaul of finite capacity: works with deadlines assume a single destination that accepts all data, and works with backhaul constraints have no per-request deadlines. Moreover, the methods that count served requests design on a deterministic channel, whereas the PrLoS channel makes the delivered count a random quantity. This paper addresses both gaps. As the strongest available baseline, we adopt the half-duplex algorithm of Tran et al. [2]: its objective is the closest to the number of timely alerts, it designs offline for known demand and jointly optimizes trajectory and radio resources, and it is the only one that models data passing through the UAV buffer to a ground gateway, which corresponds to our paths on which the UAV drops data into the ground network.

## 3. System Model

As illustrated in Figure 1, we consider a post-disaster network in which a single UAV complements a damaged ground relay network in forwarding alerts to a rescue center. Table 2 summarizes the main notation.

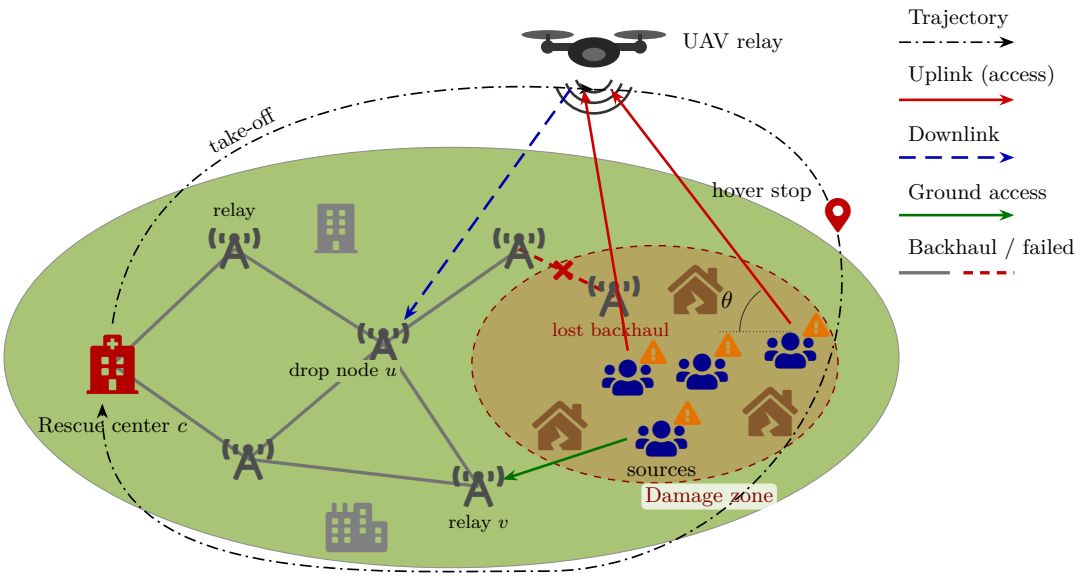



**Figure 1.** Model of the UAV-assisted alert delivery system.

### 3.1. Network and Alert Model

The post-disaster ground relay network is represented by a graph $G = (V, E)$ taken from Topology Zoo [5] and normalized into a square region of side $L_{\mathrm{box}}$ with the same scaling factor on both axes. Node $c \in V$ is the rescue center and all other nodes are relays. A surviving edge $e \in E$ has

backhaul capacity $C_e$, and a failed edge has capacity 0. Alerts are raised at a set of sources $S$, which do not belong to $G$ and can transmit only over radio links. Each alert $a \in A$ has a source $s_a \in S$, a release slot $r_a$, a deadline $d_a$, and a size of $L_a$ bits. The mission consists of $N$ slots of length $\Delta$, and each slot is split into an access phase of length $\tau\Delta$, in which sources transmit to relays or to the UAV, and a downlink phase of length $(1-\tau)\Delta$, in which the UAV transmits to nodes of $G$.

Each alert has at most $K$ candidate paths of two types: a ground path $s_a \to v \rightsquigarrow c$ through a relay $v$ in range, and a UAV path $s_a \to \text{UAV} \to u \rightsquigarrow c$ with drop node $u \in V$. The segment $v \rightsquigarrow c$ is a minimum-hop backhaul path over the surviving edges. The candidate set consists of at most two ground paths with the lowest estimated cost, complemented by the UAV paths with the lowest cost; it is fixed before planning and shared by the baselines, the bound of Section 4, and the initial plan of the proposed search. A plan assigns each alert a path $\pi_a$ or leaves it unserved; the proposed search may assign any loop-free route of either type over surviving edges, including routes outside the candidate set.

**Table 2.** Main notation.

| Symbol | Description |
|---|---|
| $G = (V, E)$, $c$ | Ground relay network and rescue center |
| $C_e$ | Backhaul capacity of edge $e$ (zero if failed) |
| $S$, $A$ | Set of sources and set of alerts |
| $s_a$, $r_a$, $d_a$, $L_a$ | Source, release slot, deadline, and size (bits) of alert $a$ |
| $N$, $\Delta$, $\tau$ | Number of slots, slot length, and access-phase fraction |
| $q[n]$, $H$, $V_{\max}$ | UAV horizontal position, altitude, and maximum speed |
| $P_k^L[n]$ | LoS probability between the UAV and ground point $k$ in slot $n$ |
| $R(b, s)$, $B_{\text{tot}}$ | Rate with bandwidth $b$ and SNR coefficient $s$; bandwidth per phase |
| $K$, $x_{a,p}$, $\pi_a$ | Maximum number of candidate paths; candidate selection and path of alert $a$ |
| $b_{e,n}$, $w_{e,n}$ | Bandwidth and bandwidth weight of radio link $e$ in slot $n$ |
| $\omega$, $M$, $M_d$ | Channel realization; numbers of evaluation and design realizations |
| $z_{a,\omega}$, $\hat{P}_{\text{timely}}$ | Timely indicator of alert $a$ in realization $\omega$; timely-delivery ratio |
| Conn | Fraction of sources with a time-respecting path to $c$ |
| $E$ | Evaluation budget of the search |

## 3.2. UAV Mobility and Channel Model

The UAV flies at a fixed altitude $H$ along a trajectory $q[0], \ldots, q[N] \in \mathbb{R}^2$ that satisfies

$$q[0] = q_s, \qquad q[N] = q_t, \qquad \|q[n+1] - q[n]\| \le V_{\max}\Delta, \quad n = 0, \ldots, N-1, \tag{1}$$

and the representative position of slot $n$ is the midpoint $\bar{q}[n] = (q[n] + q[n+1])/2$.

The link between the UAV and a ground point $k$ at position $w_k$ follows the PrLoS model of [1]:

$$\begin{aligned} d_k[n] &= \sqrt{\|\bar{q}[n] - w_k\|^2 + H^2}, \qquad \theta_k[n] = \frac{180}{\pi}\,\operatorname{atan2}\big(H, \|\bar{q}[n] - w_k\|\big), \\ P_k^L[n] &= \frac{1}{1 + a\exp\big[-b(\theta_k[n] - a)\big]}, \end{aligned} \tag{2}$$

with channel gain $g^L = \beta_0 d^{-\alpha_L}$ under line of sight and $g^N = \mu\beta_0 d^{-\alpha_N}$ otherwise. For transmit power $p$ and noise power spectral density $N_0$, let $s = pg/N_0$ (in Hz); the rate achieved with bandwidth $b$ is

$$R(b, s) = b\log_2\Big(1 + \frac{s}{b}\Big), \qquad R(0, s) = 0. \tag{3}$$

In a channel realization $\omega$, the LoS/NLoS state of the link to point $k$ in slot $n$ is drawn independently with probability $P_k^L[n]$, evaluated on the trajectory under consideration, and all plans are compared on common random numbers. The link from a source to a ground relay has a deterministic gain $\beta_0 d^{-\alpha_G}$ and exists only if $R(B_{\text{tot}}, s) \geq R_{\min}$. Sources transmit with power $p_s$, and the UAV splits a total power $P_U$ equally among at most two active downlinks per slot, so the power budget does not grow with the problem size.

## 4. Problem Formulation

A plan consists of a trajectory $q$ that satisfies (1), a path selection $x_{a,p} \in \{0,1\}$ with $\sum_p x_{a,p} \leq 1$, which allows an alert to be left unserved, and bandwidths $b_{e,n} \geq 0$ on the radio links. Plans are computed before the mission without access to the evaluation realizations and are then evaluated on independent channel realizations. During execution, every slot $n$ must satisfy the following constraints:

1. Spectrum: the total bandwidth of the access links and that of the downlinks each do not exceed $B_{\text{tot}}$, and at most two downlinks are active.
2. Capacity: a radio link $e$ carries at most $t_e \Delta\, R(b_{e,n}, s_{e,n})$ bits, with $t_e = \tau$ for access links and $t_e = 1 - \tau$ for downlinks; a backhaul edge carries at most $\Delta C_e$ bits, shared by all alerts that traverse it.
3. Causality: the bits of alert $a$ are available at its source from the beginning of slot $r_a$; a transmission in slot $n$ uses only data held in the buffer at the beginning of slot $n$, so bits received in slot $n$ can be forwarded from slot $n+1$ on; bits travel only along the selected path, and buffers are non-negative.
4. Deadline: at the beginning of slot $d_a$, all remaining bits of $a$ are dropped.

All methods share the same scheduler, which serves each queue in earliest-deadline-first (EDF) order.

**Definition 4.1** (Timely alert)**.** Alert $a$ is timely in realization $\omega$, denoted $z_{a,\omega} = 1$, if at least $L_a$ of its bits reach the center $c$ no later than the end of slot $d_a - 1$; otherwise $z_{a,\omega} = 0$.

Let $D_{a,\omega}$ be the number of bits of $a$ that reach $c$ before the deadline. Plans are compared in lexicographic order of

$$\Big(\hat{P}_{\text{timely}},\ \text{Conn},\ -\sum_{\omega}\sum_{a}\frac{L_a - D_{a,\omega}}{L_a}\Big), \qquad \hat{P}_{\text{timely}} = \frac{1}{M|A|}\sum_{\omega=1}^{M}\sum_{a\in A} z_{a,\omega}, \tag{4}$$

where Conn is the fraction of sources with a time-respecting path to $c$ on the time-expanded graph: every transmission hop takes one slot, waiting in a buffer is allowed, and a UAV link is usable in slot $n$ if $R(B_{\text{tot}}, s_{e,n}) \geq R_{\min}$. Since the key is evaluated on random channel realizations, the planning problem targets its expectation over the channel distribution:

$$\begin{aligned} \text{(P1)} \qquad & \operatorname*{lex\,max}_{q,\,x,\,b} \quad \mathbb{E}_\omega\Big[\Big(\tfrac{1}{|A|}\textstyle\sum_a z_{a,\omega},\ \text{Conn}_\omega,\ -\sum_a \frac{L_a - D_{a,\omega}}{L_a}\Big)\Big] \\ & \text{s.t.} \quad (1), \quad \textstyle\sum_p x_{a,p} \leq 1, \quad x_{a,p} \in \{0,1\}, \quad b_{e,n} \geq 0, \quad \text{and constraints 1–4,} \end{aligned} \tag{5}$$

and (4) is its empirical counterpart on $M$ evaluation realizations. Problem (P1) combines binary path choices, a non-convex dependence on the trajectory through the PrLoS probabilities, and an objective that counts discrete events, so standard convex methods do not apply directly.

It is worth noting that we adopt the timely-delivery ratio as the primary objective rather than the max–min average rate of [1], for two reasons tied to the rescue setting. First, an alert

has value only if it arrives complete before its deadline, so a design that raises average rates while delivering each alert partially or late does not help the rescue operation. Second, a max–min objective is held at zero by a single alert that cannot be delivered under any plan, and it then no longer distinguishes between plans; we therefore evaluate a leximin version of this objective and quantify its cost experimentally. Connectivity enters as the secondary objective because it measures whether each source can reach the center at all, independently of the particular alerts in the workload.

*Remark* 4.2. The third component of (4) measures the shortfall, that is, the fraction of data still missing at the deadline, rather than the total delay, because late alerts are dropped and therefore have no completion time from which a delay could be computed.

Before designing an algorithm, we establish why Problem (P1) calls for a heuristic and how its solutions can be judged: the problem is NP-hard even when the trajectory is fixed, and an upper bound is available against which any plan can be measured.

## 4.1. NP-Hardness

**Proposition 4.3.** *Maximizing the number of timely alerts is NP-hard, even when the trajectory is fixed, data can be split arbitrarily across slots, the network has only two relays, and every alert has two candidate paths.*

*Proof.* We reduce from PARTITION: given positive integers $s_1, \dots, s_m$ that sum to $2Q$, decide whether some subset sums to $Q$. Build a network with center $c$, two relays $v_1, v_2$, and two backhaul edges $(v_j, c)$ with $\Delta C_{(v_j,c)} = Q$ bits. A single source lies within range of both relays, with a gain large enough that $\tau \Delta R(B_{\text{tot}}/2, s) \geq 2Q$. Alert $i$ has $L_i = s_i$, $r_i = 0$, $d_i = 2$, and two candidates: via $v_1$ or via $v_2$. By the causality convention, the bits of a timely alert must be sent to the relay in slot 0 and cross the backhaul in slot 1. If a subset with sum $Q$ exists, route the corresponding alerts via $v_1$ and the rest via $v_2$, and allocate $B_{\text{tot}}/2$ to each access link in slot 0: each access link carries up to $2Q$ bits and each backhaul edge carries exactly $Q$ bits in slot 1, so all $m$ alerts are timely. Conversely, if all $m$ alerts are timely, the total size of the alerts routed via $v_j$ is at most $Q$ for $j = 1, 2$, while the two parts sum to $2Q$; hence each part equals $Q$ and PARTITION has a solution. The construction takes polynomial time, so deciding whether at least $m$ alerts can be timely is NP-hard. □

*Remark* 4.4. Proposition 4.3 establishes NP-hardness in the weak sense only; we do not claim membership in NP, since bandwidth is a continuous variable, and we do not consider the case $K = 1$. The hardness stems from the requirement that each alert follow exactly one path, not from scheduling on a single link: with one link and data split across slots, the problem corresponds to preemptive single-machine scheduling $1 \,|\, r_j, \text{pmtn} \,|\, \sum U_j$, which is solvable in polynomial time [25]. The non-preemptive version $1 \,|\, r_j \,|\, \sum U_j$ is strongly NP-hard [24], but our model allows bits to be split across slots, so that result does not apply directly.

## 4.2. Per-Realization Upper Bound

Fix a trajectory $q$, the candidate set, and a realization $\omega$, so that $s_{e,n}$ is known. For hop $h$ of candidate $p$ and slot $n \in [r_a + h,\, d_a - 1]$, let $f_{a,p,h,n}$ be the number of bits crossing the hop and $B_{a,p,h,n}$ the buffer at the beginning of the slot. Choose tangent points $b_k = B_{\text{tot}} 2^{-k}$, $k = 0, \dots, 9$,

and consider the linear program

$$
\begin{aligned}
\text{(P2)} \quad \max \quad & \textstyle\sum_{a\in A} z_a && && \text{(6a)}\\
\text{s.t.} \quad & \textstyle\sum_{p} x_{a,p} \le 1, && && \text{(6b)}\\
& B_{a,p,0,r_a} = L_a x_{a,p}, \qquad B_{a,p,h,r_a+h} = f_{a,p,h-1,r_a+h-1} \ (h \ge 1), && && \text{(6c)}\\
& B_{a,p,h,n+1} = B_{a,p,h,n} - f_{a,p,h,n} + f_{a,p,h-1,n}, \qquad f_{a,p,h,n} \le B_{a,p,h,n}, && && \text{(6d)}\\
& L_a z_a \le \textstyle\sum_p \sum_n f_{a,p,H_p-1,n}, && && \text{(6e)}\\
& \textstyle\sum_{(a,p,h)\,\text{via}\,e} f_{a,p,h,n} \le t_e \Delta\big(R(b_k, s_{e,n}) + R'(b_k, s_{e,n})(b_{e,n} - b_k)\big) \quad \forall k, && && \text{(6f)}\\
& \textstyle\sum_{(a,p,h)\,\text{via}\,e} f_{a,p,h,n} \le \Delta C_e, && && \text{(6g)}\\
& \textstyle\sum_{e\,\text{access}} b_{e,n} \le B_{\text{tot}}, \qquad \sum_{e\,\text{downlink}} b_{e,n} \le B_{\text{tot}}, && && \text{(6h)}\\
& x, z \in [0,1], \qquad f, B, b \ge 0, && && \text{(6i)}
\end{aligned}
$$

where $f_{a,p,-1,n} \equiv 0$, $H_p$ is the number of hops of candidate $p$, (6d) applies when $n+1 \le d_a - 1$, (6f) applies to radio links with $s_{e,n} > 0$, the flow on a radio link with $s_{e,n} = 0$ is fixed to zero, and (6g) applies to backhaul edges. Let $\mathrm{UB}_\omega(q)$ denote the optimal value of Problem (P2).

**Proposition 4.5** (Upper bound)**.** *For a fixed trajectory $q$, candidate set, and realization $\omega$, $\mathrm{UB}_\omega(q)$ is no smaller than the number of timely alerts of any plan with trajectory $q$ whose paths belong to the candidate set, executed on $\omega$.*

*Proof.* For fixed $s > 0$, $R(b,s) = b\,\phi(1/b)$ with $\phi(t) = \log_2(1+st)$ concave, so $R(\cdot, s)$ is the perspective of a concave function and is therefore concave on $b > 0$ [26]. A concave function lies below each of its tangents, hence $R(b,s) \le R(b_k,s) + R'(b_k,s)(b - b_k)$ for every $k$. Consider any plan with trajectory $q$ and its execution on $\omega$. Set $x_{a,p} = 1$ for the selected path, $z_a = z_{a,\omega}$, $f_{a,p,h,n}$ to the number of bits sent over hop $h$ in slot $n$, $B_{a,p,h,n}$ to the buffer at the beginning of the slot, and $b_{e,n}$ to the allocated bandwidth. The causality and buffer constraints yield (6c)–(6d); Definition 4.1 yields (6e); the spectrum and capacity constraints yield (6h) and (6g); and the tangent inequality yields (6f). This point is feasible and its objective value equals the number of timely alerts, which therefore cannot exceed $\mathrm{UB}_\omega(q)$. □

Problem (P2) further relaxes the integrality of $x$ and $z$, the EDF order, the limit of two downlinks, and the causality of channel knowledge, since the bound knows the realization in advance; each relaxation only enlarges the feasible region. The bound holds only for the given trajectory and candidate set and is not a global bound over all trajectories or all paths. The proposed search may route alerts outside the candidate set, so for SBS and its variants the bound refers to the problem restricted to the candidate set: a gap close to zero or negative means that the search has matched or exceeded what any plan within the candidate set could achieve on that trajectory. For each scenario, the bound is averaged over realizations and divided by $|A|$, and the gap of a method is $(\overline{\mathrm{UB}} - \overline{P})/\overline{\mathrm{UB}}$. Each method that uses the UAV is compared with the bound on its own trajectory, and the baseline without a UAV is compared with the bound of the same problem restricted to ground candidates.

To measure the tightness of the bound, we solve on reduced instances the MILP obtained from Problem (P2) by restoring $x_{a,p}, z_a \in \{0,1\}$, which we denote Problem (P3), with optimal value $\mathrm{UB}_\omega^{\mathrm{MILP}}(q)$. We then build a plan from the solution of (P3), using its selected paths and bandwidths and keeping the two largest downlinks whenever the limit is exceeded, and execute this plan on the same $\omega$, which gives the chain

$$
P_\omega(\text{MILP plan}) \ \le \ \mathrm{OPT}_\omega(q) \ \le \ \mathrm{UB}_\omega^{\mathrm{MILP}}(q) \ \le \ \mathrm{UB}_\omega(q). \tag{7}
$$

The MILP plan knows the channel in advance, so it serves only to measure the tightness of the bound and is not compared with the other methods.

Finally, we consider the computational cost of evaluating plans and computing the bound. Generating candidate paths requires one breadth-first search from $c$, which takes $O(|V| + |E|)$ time, and ranking them takes $O(|A||V| \log |V|)$. Evaluating a plan on $M$ realizations costs $O(MN(|\mathcal{L}| + |A| \log |A|))$, where $\mathcal{L}$ is the set of links, and computing Conn costs $O(MN(|V| + |E| + |S||V|))$. The evaluator records every transmission in a ledger, on which an independent checker verifies flow conservation, non-negative buffers, and the absence of future information at the same cost. Problem (P2) has $\sum_a \sum_p H_p(d_a - r_a)$ flow variables and as many buffer variables, at most $|\mathcal{L}_\mathrm{r}|N$ bandwidth variables, and at most $10|\mathcal{L}_\mathrm{r}|N$ tangent constraints, where $\mathcal{L}_\mathrm{r}$ is the set of radio links of the candidate set. We solve the bound with HiGHS through `scipy` and, instead of stating a polynomial bound for the solver, report its measured solve times with the numerical results.

## 5. Proposed Algorithm

This section presents the proposed sample-scored block search, which we denote SBS, and its schedule-repair variant SBS-R. Both methods take a scenario as input and output a plan $(q, \pi, w)$: a trajectory, a path $\pi_a$ for every alert (or none), and bandwidth weights. The candidate set of Section 3.1 only provides the initial plan; during the search, paths are generated from the current plan and may leave the candidate set. Figure 2 summarizes the pipeline. We first explain why the blocks are not solved as convex subproblems, then introduce the two ingredients shared by all blocks, namely the weighted bandwidth rule and the design scorer, and finally describe the blocks of SBS, the repair loop of SBS-R, and a leximin variant used to isolate the effect of the objective.

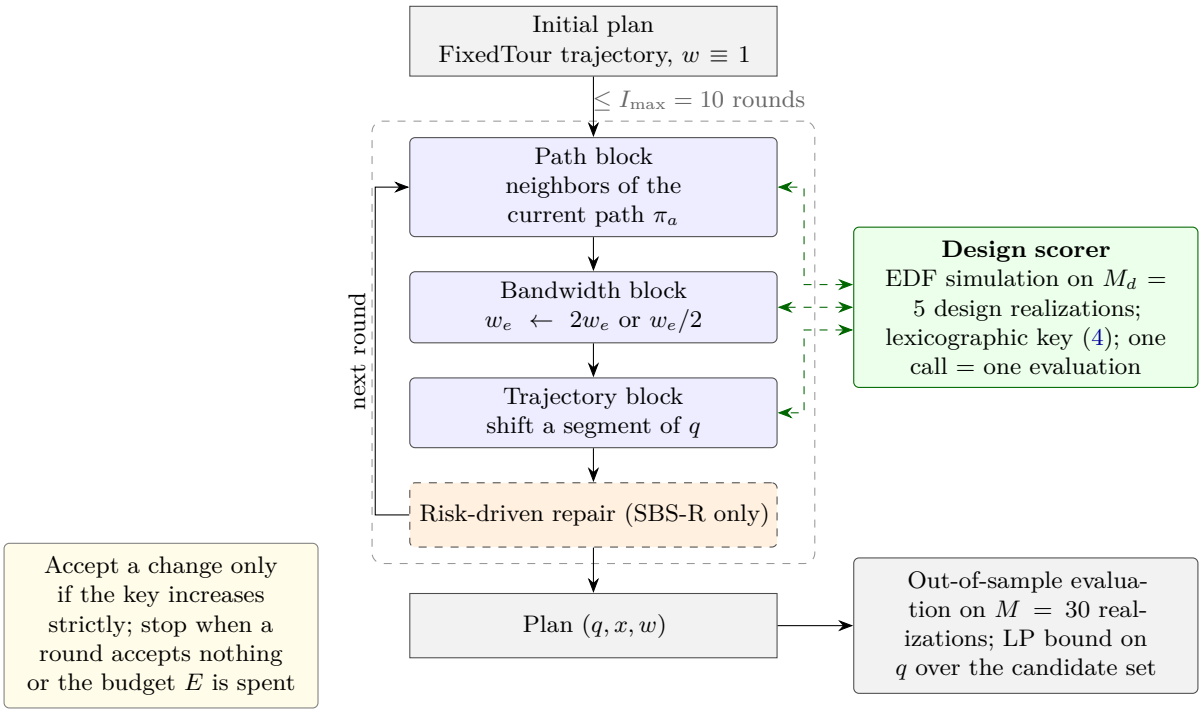



**Figure 2.** Pipeline of the proposed block search.

## 5.1. Design Rationale

We keep the three-block structure of [1] but do not solve each block as a convex problem, a choice motivated by three preliminary tests on six evaluation scenarios with ten realizations each. When the bandwidths of the solution of Problem (P2) were used as a static schedule, the timely-delivery ratio fell from 0.410 for the fixed-trajectory baseline FixedTour to 0.128; used as bandwidth-sharing weights, they reached 0.404, and paths taken from the solution of (P2) reached 0.412. Guidance from the LP therefore does not carry over to the EDF scheduler, and under the PrLoS channel the trajectory block admits no valid convex surrogate for an objective that counts alerts. Search scored by the evaluator itself, by contrast, raised the ratio to 0.427 when scoring on expected rates and to 0.449 when scoring on five sampled realizations, while scoring on expected rates even lowered one scenario from 0.372 to 0.305. We therefore implement each block as an evaluated local search: the block generates neighbors of the current plan by changing a small part of it, scores each neighbor, and keeps it only if the score improves. Compared with a surrogate-based decomposition, this design measures every accepted change on the exact objective and scheduler used for evaluation, and it requires only samples of the channel rather than a convex surrogate of it.

## 5.2. Plan Representation and Design Scoring

A plan carries a weight $w_{e,n} \geq 0$ for every radio link $e$ and slot $n$ instead of an explicit bandwidth schedule. In each phase of a slot, let $\mathcal{B}_n$ be the set of access links with queued data, or the set of at most two downlinks whose queue heads have the earliest deadlines. As illustrated in Figure 3, the bandwidth allocated to a link $e \in \mathcal{B}_n$ is

$$b_{e,n} = B_{\text{tot}} \frac{w_{e,n}}{\sum_{e' \in \mathcal{B}_n} w_{e',n}}, \tag{8}$$

with equal sharing when the weights sum to 0. The equal-sharing rule of the baselines is the special case in which all weights equal 1. The weights are fixed before the mission, whereas the actual bandwidth depends only on the current queues, so the policy uses no future information. Unlike a static bandwidth schedule, which falls out of step with how the EDF scheduler serves the queues, this rule follows the backlog in every slot while still letting the search shift spectrum toward specific links.

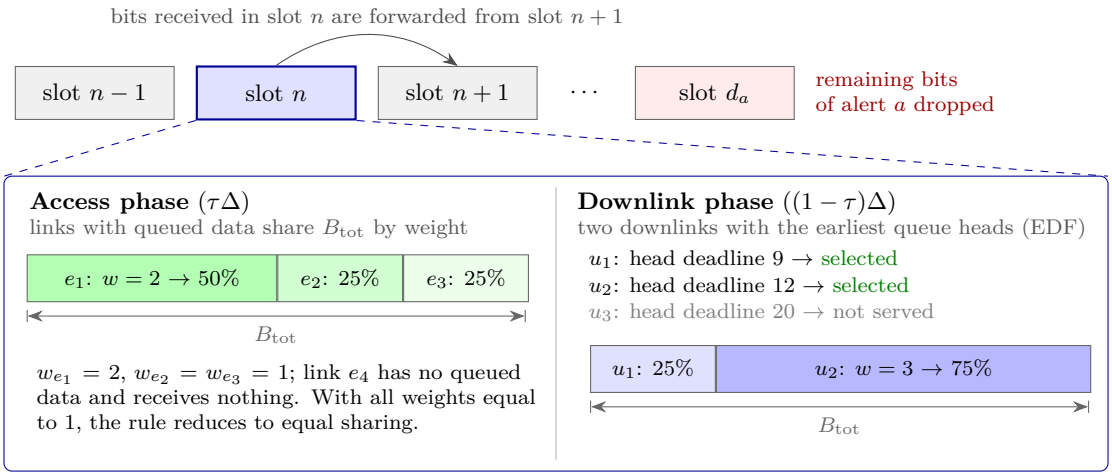



**Figure 3.** Slot structure and weighted bandwidth sharing with illustrative values.

The search never uses the evaluation realizations. It scores a plan on $M_d = 5$ design realizations, drawn from a random stream distinct from the evaluation stream, by summing the

lexicographic key (4) over these realizations. Each scoring call counts as one evaluation, and the total number of evaluations is capped by a budget $E = 2000$; when the budget is exhausted, the method returns the best plan found so far. A change is accepted only if the key increases strictly, so the key of the current plan never decreases. During design scoring, the independent ledger checker is skipped and Conn is cached per trajectory and realization, since neither affects the search decisions; this leaves the returned plan unchanged and speeds up the search by a factor of about 2.9. The final evaluation on the evaluation realizations still runs the checker.

The true objective is the expectation of the lexicographic key over the channel distribution. Summing the key over $M_d$ design realizations is a sample average approximation (SAA) of this expectation with sample size $M_d = 5$, whereas scoring on expected rates corresponds to the deterministic equivalent problem. Because design and evaluation realizations come from different random streams, all reported timely-delivery ratios are out-of-sample estimates. Our experiments on the number of design realizations examine the sample size of this SAA.

### 5.3. Block Search

SBS starts from the plan of FixedTour with all weights equal to 1 and performs at most ten rounds of three blocks. Each block changes one part of the current plan while the other two are held fixed, and every change is scored and accepted only if the key increases strictly. The path and trajectory blocks begin with one ledger evaluation of the current plan on the design realizations, which records, for every link and slot, the transmitted bits and whether the link carried its full capacity. These records make the neighbors depend on the current trajectory and bandwidth: a path is rerouted around the links that the current plan congests, and the trajectory is moved toward the radio links that are short of capacity. All random choices use one generator seeded by the scenario, so a run is reproducible.

**Path block.** The block visits the alerts in a random order drawn without replacement with probability proportional to $1/(d_a - r_a)$, so urgent alerts tend to come first while successive rounds explore different orders; with a limited budget, this order spends evaluations first where a deadline is most at risk. For alert $a$ with current path $\pi_a$, the block tries the following neighbors in turn, illustrated in Figure 4(a).

- *Kind switch:* the same drop node with the other kind of first hop, i.e., a ground path becomes a UAV path and vice versa. A ground path is generated only if the source is within ground range of the node, and only for sources that reach some relay over the ground at all, so a source isolated from the ground network keeps UAV paths only.
- *Entry switch:* the same kind of path through another entry node $v$, followed by its minimum-hop backhaul route. Each reachable node receives an estimated completion time

$$T_a(v) = T_a^{\mathrm{acc}}(v) + \sum_{e \in \mathrm{route}(v)} \Big( \frac{L_a}{\hat{C}_e} + \Delta \Big), \tag{9}$$

  where $\hat{C}_e$ is the capacity of backhaul edge $e$ minus the rate it carries in the current plan during $[r_a, d_a)$, averaged over the design realizations and floored at $C_e/1000$, and $T_a^{\mathrm{acc}}(v)$ is the ground transmission time of $L_a$ for a ground path or the mean distance from the UAV's current positions in $[r_a, d_a)$ to $v$ divided by $V_{\max}$ for a UAV path. The block draws $M_e = 3$ entry nodes without replacement with probability proportional to $1/T_a(v)$ and tries them in increasing $T_a(v)$, which favors nearby relays with spare backhaul without excluding the others.

- *Backhaul detour:* the backhaul edge $(u, v)$ of $\pi_a$ that the current plan congests in the most slots of $[r_a, d_a)$ is removed, and the path is continued from $u$ by a minimum-hop route to $c$ that avoids $(u, v)$ and the nodes of $\pi_a$ before $u$. The part of $\pi_a$ up to $u$ is kept. No detour is generated when $\pi_a$ crosses no congested edge or when no such route exists.
- *Skip:* the alert is left unserved, which frees capacity for the others.

An alert that is currently unserved tries its candidate paths instead. Every generated path is a loop-free route from $s_a$ to $c$ over surviving edges, so the neighborhood stays within the feasible set while reaching paths that are not in the candidate set.

**Bandwidth block.** The block visits the radio links in use in decreasing order of the total size of the alerts they carry and tries to multiply all weights of a link by 2, or by 1/2 if doubling is rejected. A radio link that enters the plan through a new path starts with weight 1.

**Trajectory block.** The block moves a segment $q[i], \dots, q[j]$ of the current trajectory, $1 \le i \le j \le N-1$, by a common displacement $\lambda u$ with a unit direction $u$, as illustrated in Figure 4(b). The steps inside the segment are unchanged, so constraint (1) only restricts the two boundary steps, and the largest feasible displacement is

$$\lambda^\star = \max\{\lambda \ge 0 : \ \|q[i] + \lambda u - q[i-1]\| \le V_{\max}\Delta, \ \|q[j] + \lambda u - q[j+1]\| \le V_{\max}\Delta\}, \qquad (10)$$

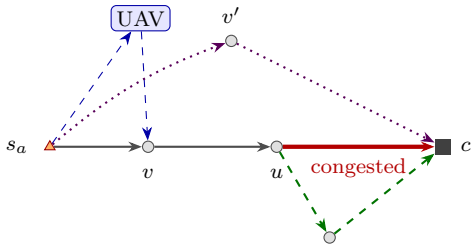
obtained in closed form from two quadratic inequalities. When $\lambda^\star = 0$ because a boundary lies on a segment flown at full speed, the segment is widened by one slot on each side, at most ten times. The segments come from the ledger evaluation: for every radio link between the UAV and a ground point $k$, the priority of slot $n$ is the number of design realizations in which the link is saturated at $n$, multiplied by $\sum_a L_a/(d_a - r_a)$ over the alerts on the link whose window contains $n$. Consecutive slots with positive priority form a segment, $u$ points from the centroid of the segment toward $k$, and the block draws $W = 5$ segments with probability proportional to their total priority. Each drawn segment is tried with $\lambda = \lambda^\star$ and then with $\lambda^\star/2$. In the example of Figure 4(b), the UAV hovers between two sources whose uplinks share the access band; if the alert of $s_1$ is larger or more urgent, its uplink accumulates more priority and the segment is pulled toward $s_1$. To keep moves that change the zone order available, the block also tries, with probability 0.1 per round, one tour drawn uniformly from a family of up to 90 feasible tours, which visit an ordered subset of the damage zones in straight lines at speed $V_{\max}$, hover at the zone centers or the centroids of their sources, and split the remaining time among the stops equally, in proportion to the number of alerts of each zone, or in proportion to $\sum_a 1/(d_a - r_a)$ over the zone. Every trajectory produced by the block satisfies (1) by construction.

The search stops when a round accepts no change or the budget is spent. Compared with SCA-based trajectory design, the trajectory block gives up the gradient of a convex surrogate, but every move it makes is feasible and evaluated on the exact objective; compared with scanning a fixed family of tours, the moves follow the incumbent and the current congestion of the plan, which lets the UAV settle at positions that no predefined tour contains.

## 5.4. Schedule Repair

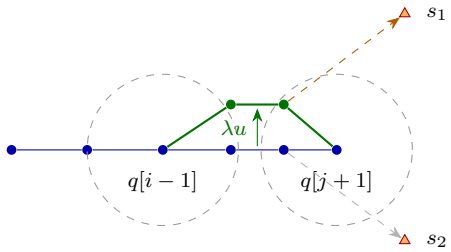
The blocks of SBS change one decision at a time without regard to which alerts are about to miss their deadlines. SBS-R targets these alerts directly by adding a repair loop after the trajectory block of every round, as illustrated in Figure 5. A ledger evaluation of the current plan records, for each alert and design realization, its slack $d_a - 1 - t_a$, where $t_a$ is the completion slot and the slack is set to $-1$ for late alerts, together with the hop that holds the alert's data for the most slots and the links whose capacity is exhausted. The risk of an alert is the fraction of design

(a) Neighbors of the path of alert $a$




current path $s_a \to v \to u \to c$ (solid); kind switch via the UAV to $v$; entry switch to $v'$ drawn by $1/T_a(v')$; detour around the congested edge from $u$; skip

(b) Shift of a trajectory segment

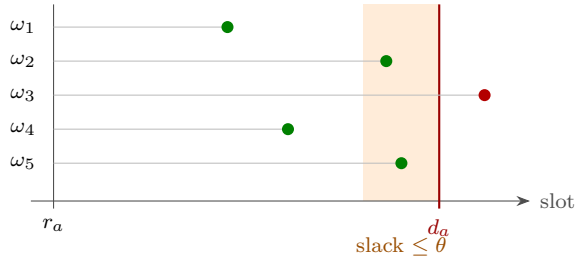



the segment $q[i..j]$ (here two points) moves by $\lambda u$ toward the source whose saturated uplink has the highest priority; the dashed circles of radius $V_{\max}\Delta$ around $q[i-1]$ and $q[j+1]$ bound $\lambda$ by (10)

**Figure 4.** Neighborhoods of the path block and the trajectory block.

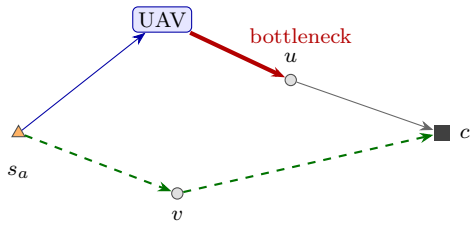
realizations in which it is late or has a slack of at most $\theta = 2$ slots, and the loop considers at most $R = 10$ alerts with the highest risk. If the bottleneck hop of such an alert is a radio link, Operator 1 doubles its weight within the slots $[r_a, d_a)$ only, which moves bandwidth from the other links of the same phase to this alert. Operator 2 generates the neighbors of the alert's path as in the path block, ranks them first by the number of congested links in the alert's window and then by estimated cost, and accepts the first one that increases the key. All operators share the budget with the three blocks, so SBS-R and SBS have the same maximum number of evaluations. The mechanism follows the destroy-and-repair principle of large neighborhood search [22].

(a) Risk detection on design realizations




completion slot $t_a$ in each design realization; risk = share of realizations that are late or have slack $d_a - 1 - t_a \leq \theta$ (here 3/5)

(b) Repair operators for an at-risk alert




Operator 1: double the weight of the bottleneck hop only within $[r_a, d_a)$; Operator 2: switch to a neighbor path, ranked by congested links, then cost

**Figure 5.** Risk-driven schedule repair in SBS-R: (a) risk detection; (b) repair operators.

## 5.5. Overall Algorithm and Complexity

Algorithm 1 summarizes the method; SBS is obtained with repair disabled and SBS-R with repair enabled.

One round uses at most $1 + (M_e + 3)|A|$ evaluations for the path block, $2|\mathcal{L}_{\text{used}}|$ for the bandwidth block, and $2 + 2W$ for the trajectory block, plus $1 + R(M_e + 3)$ for the repair loop of SBS-R, where $\mathcal{L}_{\text{used}}$ is the set of radio links in use. Generating the neighbors of one alert takes $O(|V|(|V| + |E|))$ time for the routes, and drawing the segments takes $O(N|\mathcal{L}_{\text{used}}||A|)$, both negligible next to an evaluation. Each evaluation costs $O(M_d N(|\mathcal{L}| + |A| \log |A|))$, and risk detection scans the ledger in $O(M_d(N|\mathcal{L}| + T))$, where $T$ is the number of transmission records.

**Algorithm 1** Sample-scored block search with schedule repair.

**Input:** Scenario, candidate set, design realizations, budget $E$, maximum rounds $I_{\max} = 10$, seed
**Output:** Trajectory $q$, paths $\pi$, bandwidth weights $w$

$(q, \pi) \leftarrow$ plan of FixedTour; $w \leftarrow 1$; $\kappa \leftarrow$ key of $(q, \pi, w)$
**for** $i = 1, \ldots, I_{\max}$ **do**
    Path block: ledger evaluation; for each alert in urgency-weighted random order, try its kind switch, $M_e$ entry switches drawn by $1/T_a(v)$, the detour around its most congested backhaul edge, and skipping it
    Bandwidth block: for each link in use, try $w_e \leftarrow 2w_e$, then $w_e \leftarrow w_e/2$
    Trajectory block: ledger evaluation; draw $W$ segments by priority and try each shift with $\lambda^\star$, then $\lambda^\star/2$; with probability 0.1 try one tour of the feasible tour family
    **if** repair is enabled **then**
        Ledger evaluation; select at most $R$ alerts with the highest deadline risk
        **for all** at-risk alerts $a$ **do**
            Operator 1: double the weight of the bottleneck hop within $[r_a, d_a)$
            Operator 2: try neighbor paths by number of congested links, then by cost
        **end for**
    **end if**
    Accept each trial iff the key increases strictly; stop as soon as the budget $E$ is spent
    **if** no change was accepted in this round **then break**
    **end if**
**end for**
**return** $(q, \pi, w)$

The budget caps the total number of evaluations, so the planning time of SBS, SBS-Lex, and SBS-R is $O(EM_dN(|\mathcal{L}| + |A| \log |A|))$, independent of the actual number of rounds. The method only guarantees that the key of the plan does not decrease; it does not guarantee optimality. The LP bound (P2), solved on the trajectory chosen by each method, measures the remaining room for improvement within the candidate set; since SBS may route alerts outside this set, its value on a realization can exceed the bound.

To separate the effect of the objective from that of the search, SBS-Lex runs the same search as SBS with a different key. Let $\bar{f}_a$ be the fraction of the bits of $a$ that reach the center before the deadline, averaged over the design realizations. The average delivery rate of $a$ divided by its required rate $L_a/((d_a - r_a)\Delta)$ equals exactly $\bar{f}_a$, so $\min_a \bar{f}_a$ is a normalized version of the max–min objective of [1]. Because a single undeliverable alert forces $\min_a \bar{f}_a = 0$ for every plan, SBS-Lex compares plans in leximin order:

$$(\bar{f}_{(1)}, \bar{f}_{(2)}, \ldots, \bar{f}_{(|A|)},\ \textstyle\sum_\omega \sum_a z_{a,\omega},\ \text{Conn}), \qquad \bar{f}_{(1)} \le \bar{f}_{(2)} \le \cdots \le \bar{f}_{(|A|)}, \tag{11}$$

where an unserved alert has $\bar{f}_a = 0$. The difference between SBS and SBS-Lex measures the cost of optimizing fairness instead of the number of timely alerts.

## 6. Numerical Results

This section evaluates the proposed algorithm in simulation. The experiments address four questions: whether jointly optimizing trajectory, paths, and bandwidth improves the timely-delivery ratio over partial optimization and over a strong method from the literature, which design choices produce the gain, when the UAV matters, and whether the conclusions survive parameter changes and a realistic source layout.

## 6.1. Simulation Setup

The first scenario family, called Wide, is generated deterministically from the 130 Topology Zoo topologies that have between 15 and 40 nodes, geographic coordinates, and a connected graph. After stratifying them by node count, we use ten topologies for evaluation with three replicates each and reserve three others for development with two replicates each. Alert sizes inherit their relative proportions from the demands of the SNDlib instance `abilene` [6], whereas damage zones, source positions, release times, deadlines, and failed edges are drawn from seeded random streams. The second family, called Narrow, keeps every seed and only lowers the backhaul capacity from 1000 kbit/s to 50 kbit/s, so that both families share the same topologies, alerts, and failure patterns. Table 3 lists the simulation parameters, whose channel constants follow those of [1]. All scenarios are synthesized from public data and do not represent measured rescue networks.

**Table 3.** Simulation parameters.

| Parameter | Symbol | Value |
|---|---|---|
| Simulation area side | $L_{\text{box}}$ | 2000 m |
| Evaluation / development topologies | — | 10 / 3 |
| Replicates per topology (evaluation / development) | — | 3 / 2 |
| Channel realizations | $M$ | 30 |
| Slot length / number of slots | $\Delta$, $N$ | 1 s / 150 |
| Access-phase fraction | $\tau$ | 0.5 |
| UAV altitude / maximum speed | $H$, $V_{\max}$ | 100 m / 50 m/s |
| UAV power / concurrent downlinks | $P_U$ | 0.1 W / 2 |
| Source power | $p_s$ | 0.1 W |
| Bandwidth per phase | $B_{\text{tot}}$ | 1 MHz |
| Noise power spectral density | $N_0$ | −140 dBm/Hz |
| PrLoS parameters | $(a, b)$ | (9.61, 0.16) |
| NLoS attenuation / LoS, NLoS exponents | $\mu$, $\alpha_L$, $\alpha_N$ | 0.2 / 2.2, 3.3 |
| Reference gain | $\beta_0$ | −50 dB |
| Ground path-loss exponent | $\alpha_G$ | 2.8 |
| Usable-rate threshold | $R_{\min}$ | 10 kbit/s |
| Candidate paths (max. ground paths) | $K$ | 3 (2) |
| Damage zones: count / separation / radius | — | 3 / 600 m / 400 m |
| Sources / position std. dev. | $\lvert S\rvert$, $\sigma$ | 15 / 150 m |
| Alerts / deadline window | $\lvert A\rvert$, $d_a - r_a$ | 40 / 20–60 slots |
| Reference size (clip range, instance) | $L_{\text{ref}}$ | 250 kbit ([0.25, 4], `abilene`) |
| Edge failure prob. inside / outside zones | $p_{\text{in}}$, $p_{\text{out}}$ | 0.60 / 0.05 |
| Backhaul capacity (Wide / Narrow) | $C_e$ | 1000 kbit/s / 50 kbit/s |
| LP tangents / bootstrap resamples | — | 10 / 10000 |

For the case study, the sources are instead placed at damaged buildings in the RescueNet dataset [7], which contains UAV images of Mexico Beach taken after Hurricane Michael together with segmentation masks. On 30 image–mask pairs, we extract the 8-connected regions labeled as minor damage, major damage, or total destruction, discard those smaller than 10 $\text{m}^2$, and represent each remaining region by its pixel closest to the centroid. Because the dataset does not report the ground sampling distance, we derive the pixel size from the altitude and calibrated focal length stored in the image metadata, which gives values between 1.96 and 2.46 cm, and we project every point into UTM zone 16N using the GPS position and gimbal orientation. Merging points from overlapping images that lie within 10 m of each other reduces the 68 damaged regions to 63 targets, which are then normalized into a 2 km region with the same scale on both axes.

The resulting family, called RescueNet, follows every other rule of Wide: it uses the same ten evaluation topologies, three replicates, and random streams for alerts and failed edges, but

every target becomes a source and the three damage zones are obtained by k-means clustering of the targets. Since the targets spread over the whole region rather than clustering around three centers, only 0.376 of the sources have a ground relay in range, compared with 0.509 in Wide. All targets come from a single geographic layout observed in overlapping images, so we report the case-study results descriptively.

We compare the proposed methods SBS and SBS-R with the following benchmark schemes:

- NoUAV: only ground paths are used.
- FixedTour: the UAV follows a fixed tour of the damage zones, and paths are chosen per alert.
- SBS-Lex: the search of SBS with the fairness key (11).
- Tran-IA: the inner approximation algorithm of Tran et al. [2], adapted to our model as described below.

NoUAV and FixedTour share the candidate set, the EDF scheduler, and equal bandwidth sharing, and both plan from expected rates. NoUAV routes each alert over its lowest-cost ground path and leaves an alert unserved when its source has no relay in range, at a planning cost of $O(|A|K)$. FixedTour flies the UAV to the zone centers in nearest-neighbor order with equal hover times and assigns each alert to the candidate that completes earliest when simulated alone on this trajectory; an alert is left unserved if no candidate meets its deadline. Its planning cost is $O(|A|KNH_{\max})$, where $H_{\max}$ is the largest number of hops of a candidate.

Tran-IA, the half-duplex algorithm of Tran et al. [2], maximizes the number of devices whose data are uploaded to the UAV within an upload window and then offloaded to a gateway, with a binary variable $\lambda_k$ indicating whether device $k$ is served. The algorithm relaxes $\lambda_k$ to $[0, 1]$ and adds the penalty $\mu \sum_k \lambda_k(\lambda_k - 1)$; each iteration then linearizes the penalty, replaces the rates by concave lower bounds around the current point, and solves a convex problem in trajectory, bandwidth, and $\lambda$. Table 4 lists the changes required by our model. They preserve the structure of the algorithm but may put it at a disadvantage, because the original work models neither a finite backhaul nor an EDF scheduler.

**Table 4.** Adaptation of Tran-IA to our model.

| **Original paper** | **This work** |
|---|---|
| Device $k$ with $S_k$ bits, a start time, and an upload deadline | Alert $a$ with $L_a$ bits, release slot $r_a$, and upload window $[r_a,\, r_a + \lceil \theta W_a \rceil)$ |
| One gateway; offloading from the upload deadline until the end of the mission | Gateway is the entry node of the lowest-cost candidate; offloading within $[r_a + \lceil \theta W_a \rceil,\, d_a - b_a)$ |
| Device and UAV powers optimized | Fixed powers, as for every method |
| Rician channel with path loss $\omega_0 d^{-\alpha}$ | Power law $\beta d^{-\alpha}$ fitted to the expected PrLoS gain |
| Finite UAV memory | None, since the UAV buffer is unlimited in our model |
| No limit on concurrent downlinks | At most two downlinks with the largest bandwidths kept per slot |
| Feasibility problem for initialization | Initialized from the FixedTour trajectory, equal bandwidths, and $\lambda = 0$, which is always feasible |
| No values given for $\mu$ and the convergence threshold | $\theta$ and $\mu$ selected on the development set; stop when the objective changes by less than $10^{-3}$ or after 30 iterations |

In the adapted version, each alert plays the role of a device, and its gateway is the node where its lowest-cost candidate enters the ground network. Given $b_a$ slots of backhaul delay and $W_a = d_a - b_a - r_a$, the upload window is $[r_a,\, r_a + \lceil \theta W_a \rceil)$ and the offloading window ends at $d_a - b_a$, which leaves enough time for the data to cross the backhaul before the deadline. The Rician channel of the original work is replaced by a power law $\beta d^{-\alpha}$ fitted by least squares to the expected PrLoS gain; on the development scenarios, the fitted exponent lies between 3.34 and

3.42 with $R^2$ of at least 0.9936. Each iteration solves a second-order cone program with `cvxpy` and Clarabel. The algorithm starts from the FixedTour trajectory with equal bandwidths and $\lambda = 0$, stops when the relaxed objective changes by less than $10^{-3}$ or after 30 iterations, and converts its solution into a static bandwidth schedule that keeps at most two downlinks per slot. We selected $\theta = 0.333$ and $\mu = 10$ on 6 development scenarios from the grid $\theta \in \{1/3, 1/2, 2/3\}$, $\mu \in \{1, 10\}$, although all grid cells were within 0.041 of each other in timely-delivery ratio.

Each method plans once per scenario, and its plan is evaluated on 30 channel realizations with common random numbers. The development topologies were used only in two preliminary runs: one confirmed $E = 2000$ and $M_d = 5$, and one fixed the parameters of the neighborhoods, $M_e = 3$ entry nodes, $W = 5$ segments, and a catalogue-tour probability of 0.1. For the second run, we fixed the decision rule in advance: a default is replaced only if another value beats it by more than half the width of its bootstrap CI. No alternative came close; the largest paired gain over the default was 0.012 on six scenarios, about twice the spread of 0.007 between three random seeds. No parameter was tuned on the evaluation set. The main experiment runs NoUAV, FixedTour, SBS, SBS-Lex, SBS-R, Tran-IA, and eight ablation variants of SBS-R on the 30 evaluation scenarios of each family, and it solves the LP bound for every realization on the trajectory of each method. For each realization, the envelope bound is the largest of the bounds on the trajectories of FixedTour, SBS, SBS-Lex, and SBS-R, and the envelope gap of a method is measured against it in the same way as the gap to its own bound. The result tables report planning times in seconds and the mean number of evaluations used by the search. Two further experiments vary the budget $E$ from 250 to 4000 on Wide and vary one parameter at a time on the first replicate of each evaluation topology.

Results are averaged first over realizations and then over scenarios. Because the replicates of a topology are not independent, 95% confidence intervals (CIs) are computed with a bootstrap clustered by topology, and paired comparisons are based on the ten topology-level differences. Each comparison uses an exact sign-flip test over all $2^{10}$ sign assignments, a bootstrap CI with 10 000 resamples, and the matched-pairs rank-biserial correlation as effect size, and its $p$-value is Holm-corrected within its group, namely the eight comparisons among our methods or the four comparisons against Tran-IA. Ablation, sensitivity, and case-study results are reported descriptively. All bound LPs were solved to optimality with HiGHS. The implementation uses Python 3.12.3, `scipy` 1.18.1, HiGHS 1.12.0, `cvxpy` 1.9.2, and Clarabel 0.11.1, runs 12 worker processes on a 12-CPU machine, and consumed 95.9 compute hours for all experiments together.

## 6.2. Comparison with Benchmark Schemes

**Table 5.** Main comparison on both scenario families.

| Backhaul | Method | Timely ↑ | 95% CI | Conn ↑ | Own gap ↓ | Env. gap ↓ | Plan. (s) | Eval. |
|---|---|---|---|---|---|---|---|---|
| 1000 kbit/s | NoUAV | 0.413 | [0.319, 0.487] | 0.509 | 0.171 | 0.428 | 0.0 | 1 |
| 1000 kbit/s | FixedTour | 0.494 | [0.391, 0.582] | 1.000 | 0.238 | 0.305 | 0.2 | 1 |
| 1000 kbit/s | SBS | 0.559 | [0.452, 0.648] | 1.000 | 0.191 | 0.204 | 172.3 | 1861 |
| 1000 kbit/s | SBS-Lex | 0.369 | [0.283, 0.445] | 1.000 | 0.471 | 0.492 | 206.8 | 1950 |
| 1000 kbit/s | SBS-R | 0.560 | [0.448, 0.656] | 1.000 | 0.190 | 0.205 | 181.1 | 1975 |
| 50 kbit/s | NoUAV | 0.370 | [0.283, 0.437] | 0.509 | 0.229 | 0.450 | 0.0 | 1 |
| 50 kbit/s | FixedTour | 0.430 | [0.344, 0.500] | 1.000 | 0.287 | 0.350 | 0.2 | 1 |
| 50 kbit/s | SBS | 0.517 | [0.423, 0.598] | 1.000 | 0.199 | 0.213 | 212.6 | 1941 |
| 50 kbit/s | SBS-Lex | 0.336 | [0.260, 0.402] | 1.000 | 0.484 | 0.503 | 248.4 | 1969 |
| 50 kbit/s | SBS-R | 0.521 | [0.425, 0.601] | 1.000 | 0.191 | 0.204 | 214.8 | 2001 |

As Table 5 shows, jointly optimizing the plan pays off clearly over both baselines. On Wide, SBS delivers 0.559 of the alerts on time, against 0.494 for FixedTour and 0.413 for NoUAV, and

it outperforms NoUAV on 30 and FixedTour on 29 of the 30 scenarios. The repair variant SBS-R reaches 0.560, practically the same as SBS: it is better on 16 scenarios, equal on 4, and worse on 10. SBS-Lex reaches only 0.369, below even FixedTour, although every method that uses the UAV brings connectivity close to one. The path neighborhoods are used: in the final plans of SBS and SBS-R, on average 5.7 and 5.3 of the 40 alerts per scenario follow a route that is not among their $K = 3$ candidates, and 7.7 and 8.4 under narrow backhaul, where rerouting around congested edges matters more. Reducing the backhaul to 50 kbit/s lowers all ratios but leaves the ranking unchanged, with SBS at 0.517, SBS-R at 0.521, FixedTour at 0.430, and NoUAV at 0.370.

The relative gaps to the upper bounds point in the same direction. Because SBS may leave the candidate set, its gap refers to the problem restricted to that set and can be negative on single realizations, yet on average it stays clearly positive. Exceeding the restricted bound happens on 114 of the 5400 realizations of SBS and SBS-R in the main experiment and the case study. For each of them we replanned, reproduced the recorded count, and solved Problem (P2) again on the same trajectory with the candidate set extended by the plan's own paths, which makes the bound valid for that plan; every count stayed below this extended bound, by at least 1.09 alerts. The gap of SBS to the bound on its own trajectory is 0.191, whereas that of FixedTour to the bound on the FixedTour trajectory is 0.238, because SBS delivers more alerts and also chooses trajectories with higher bounds. Relative to the envelope bound of 0.678, the gaps are 0.204 for SBS and 0.305 for FixedTour. SBS-Lex shows a gap of 0.471 even though its trajectory has nearly the same bound as that of SBS, which indicates that its loss stems from the path and bandwidth choices made under the leximin objective rather than from the trajectory.

**Table 6.** Paired comparisons with the baselines.

| Family | Comparison | Difference | 95% CI | $r$ | $p$ | $p_{\text{Holm}}$ |
|---|---|---|---|---|---|---|
| Wide | SBS-R − SBS | 0.002 | [−0.008, 0.015] | 0.02 | 0.8477 | 0.8477 |
| Wide | SBS − FixedTour | 0.064 | [0.044, 0.086] | 1.00 | 0.0020 | 0.0156 |
| Wide | SBS − SBS-Lex | 0.190 | [0.157, 0.226] | 1.00 | 0.0020 | 0.0156 |
| Wide | SBS-R − FixedTour | 0.066 | [0.046, 0.088] | 1.00 | 0.0020 | 0.0156 |
| Narrow | SBS-R − SBS | 0.003 | [−0.004, 0.010] | 0.31 | 0.3984 | 0.7969 |
| Narrow | SBS − FixedTour | 0.087 | [0.065, 0.112] | 1.00 | 0.0020 | 0.0156 |
| Narrow | SBS − SBS-Lex | 0.181 | [0.145, 0.220] | 1.00 | 0.0020 | 0.0156 |
| Narrow | SBS-R − FixedTour | 0.091 | [0.070, 0.112] | 1.00 | 0.0020 | 0.0156 |

The paired tests in Table 6 confirm that these differences are systematic. SBS exceeds FixedTour by 0.064 on average on Wide, with a 95% CI of [0.044, 0.086], and by 0.087 on Narrow, while SBS-R exceeds FixedTour by 0.066 and 0.091, respectively. These comparisons and SBS − SBS-Lex all reach $p_{\text{Holm}} = 0.0156$, the smallest value attainable with ten topologies and eight comparisons, because all ten topologies agree in sign and the rank-biserial correlation equals one. The difference between SBS-R and SBS, in contrast, is only 0.002 on Wide, with a CI of [−0.008, 0.015] that contains zero, and 0.003 on Narrow, with $p_{\text{Holm}} = 0.8477$ and 0.7969, respectively; the data therefore provide no evidence that schedule repair improves on SBS.

**Table 7.** Results of Tran-IA.

| Backhaul | Method | Timely ↑ | 95% CI | Gap ↓ | Planning (s) ↓ |
|---|---|---|---|---|---|
| 1000 kbit/s | Tran-IA | 0.530 | [0.462, 0.590] | 0.304 | 234.2 |
| 50 kbit/s | Tran-IA | 0.438 | [0.393, 0.478] | 0.401 | 222.9 |

Table 7 reports the results of Tran-IA, which can be read against those of FixedTour, SBS, and SBS-R in Table 5. On Wide, Tran-IA delivers 0.530 of the alerts on time, more than FixedTour at 0.494 but fewer than SBS at 0.559 and SBS-R at 0.560. Under narrow backhaul, its ratio of 0.438 falls close to that of FixedTour at 0.430 and clearly below that of SBS at 0.517. Tran-IA also lies farther from the bound on its own trajectory, with gaps of 0.304 and 0.401, and it plans longer than SBS-R, taking 234.2 instead of 181.1 seconds per scenario on Wide.

**Table 8.** Paired comparisons with Tran-IA.

| Family | Comparison | Difference | 95% CI | $r$ | $p$ | $p_{\text{Holm}}$ |
|---|---|---|---|---|---|---|
| Wide | SBS-R − Tran-IA | 0.030 | [−0.020, 0.073] | 0.49 | 0.2480 | 0.4570 |
| Wide | SBS − Tran-IA | 0.028 | [−0.014, 0.066] | 0.56 | 0.2285 | 0.4570 |
| Narrow | SBS-R − Tran-IA | 0.083 | [0.024, 0.132] | 0.75 | 0.0273 | 0.1094 |
| Narrow | SBS − Tran-IA | 0.079 | [0.020, 0.128] | 0.75 | 0.0293 | 0.1094 |

Table 8 quantifies these differences. On Wide, SBS-R exceeds Tran-IA by 0.030 on average, with a CI of [−0.020, 0.073] and $p_{\text{Holm}} = 0.4570$; on Narrow, the advantage grows to 0.083, with a CI of [0.024, 0.132] and $p_{\text{Holm}} = 0.1094$, and SBS behaves similarly. Under narrow backhaul, the bootstrap CI excludes zero and the rank-biserial correlation reaches 0.75, yet no difference remains significant at the 5% level after Holm correction for the four comparisons. This pattern is consistent with the design of Tran-IA, which plans on a deterministic effective channel with a static bandwidth schedule and without knowledge of the backhaul capacity, and therefore loses most of its advantage when the backhaul is narrow.

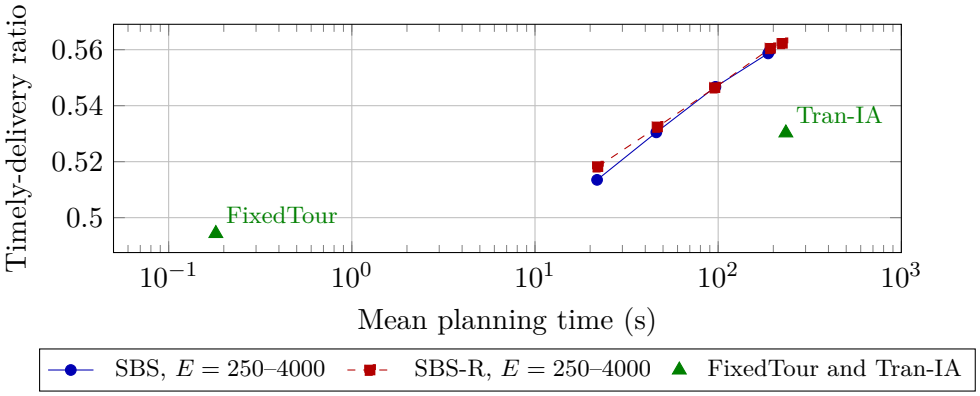



**Figure 6.** Timely-delivery ratio versus planning time on Wide.

Figure 6 places Tran-IA and FixedTour on the quality–runtime curves that SBS and SBS-R trace for budgets from 250 to 4000. Already with a budget of $E = 500$, SBS reaches 0.530, the level of Tran-IA, at a fraction of its planning time, and larger budgets move both SBS and SBS-R toward the saturation level examined below.

## 6.3. Ablation and Algorithm Analysis

The ablation in Table 9 identifies where the gain comes from, with $\Delta$ denoting the topology-level mean difference in timely-delivery ratio between a variant and SBS-R. Disabling the path block changes the timely-delivery ratio of SBS-R by −0.045 on Wide and −0.068 on Narrow, and disabling the trajectory block changes it by −0.041 and −0.032. Designing on expected rates instead of five sampled realizations changes it by −0.038 and −0.045, even though it plans much faster. The remaining repair and bandwidth variants, which keep only one repair operator, repair without backlog information, or skip the bandwidth block, stay within about one hundredth of

**Table 9.** Ablation of SBS-R.

| | Wide | | Narrow | |
|---|---|---|---|---|
| **Variant** | **Timely ↑** | **Δ vs. SBS-R [CI]** | **Timely ↑** | **Δ vs. SBS-R [CI]** |
| Full SBS-R | 0.560 | — | 0.521 | — |
| Operator 1 only | 0.560 | −0.001 [−0.014, 0.009] | 0.516 | −0.005 [−0.012, 0.001] |
| Operator 2 only | 0.562 | 0.001 [−0.008, 0.011] | 0.521 | 0.001 [−0.008, 0.010] |
| Repair without backlog | 0.558 | −0.002 [−0.007, 0.003] | 0.522 | 0.001 [−0.006, 0.009] |
| No path block | 0.515 | −0.045 [−0.067, −0.027] | 0.453 | −0.068 [−0.091, −0.048] |
| No trajectory block | 0.519 | −0.041 [−0.055, −0.029] | 0.489 | −0.032 [−0.041, −0.024] |
| No bandwidth block | 0.564 | 0.003 [−0.005, 0.013] | 0.518 | −0.003 [−0.012, 0.006] |
| Expected-rate design | 0.523 | −0.038 [−0.053, −0.024] | 0.476 | −0.045 [−0.052, −0.037] |
| Catalogue trajectory block | 0.567 | 0.006 [−0.014, 0.024] | 0.528 | 0.008 [−0.006, 0.022] |

SBS-R. The gain of the search therefore comes from selecting paths and trajectory jointly on sampled realizations, while the bandwidth weights and the repair operators matter little once the EDF scheduler shares bandwidth according to the backlog.

The variant with the catalogue trajectory block replaces the local segment shifts by a scan of every tour in the family of feasible tours in each round, the design of an earlier version of this method. It changes the ratio by 0.006 on Wide and 0.008 on Narrow, and both CIs in Table 9 contain zero. Local moves around the current trajectory thus perform on par with a full scan of predefined tours while spending about 12 instead of up to 90 evaluations per round. The small positive mean difference nonetheless hints that occasional large moves, such as changing the order in which the zones are visited, retain some value beyond the 10% chance of trying one catalogue tour per round, which suggests mixing the two move types more evenly as future work.

**Table 10.** Results by ground connectivity.

| **Family** | **Ground-reachable** | **Scenarios** | **Conn NoUAV** | **NoUAV** | **FixedTour** | **SBS** | **SBS − NoUAV** |
|---|---|---|---|---|---|---|---|
| Wide | At most 1/3 | 10 | 0.260 | 0.230 | 0.276 | 0.334 | 0.104 |
| Wide | 1/3 to 2/3 | 11 | 0.527 | 0.420 | 0.514 | 0.583 | 0.162 |
| Wide | Above 2/3 | 9 | 0.763 | 0.608 | 0.713 | 0.779 | 0.170 |
| Narrow | At most 1/3 | 10 | 0.260 | 0.210 | 0.260 | 0.328 | 0.118 |
| Narrow | 1/3 to 2/3 | 11 | 0.527 | 0.398 | 0.456 | 0.542 | 0.144 |
| Narrow | Above 2/3 | 9 | 0.763 | 0.514 | 0.588 | 0.698 | 0.184 |

To see when the UAV matters, Table 10 groups the scenarios by the fraction of sources that have a ground relay in range. Without a UAV, both the connectivity, listed as Conn NoUAV, and the timely-delivery ratio closely follow this fraction, the latter with correlation coefficients of 0.90 on Wide and 0.87 on Narrow, since alerts from sources without a relay in range cannot be delivered at all. The UAV raises connectivity close to one in every group, yet the gain of SBS over NoUAV does not shrink where most sources already reach a ground relay: on Wide it grows from 0.104 in the lowest group to 0.162 and 0.170, and on Narrow from 0.118 to 0.184. Where sources reach the ground network, the search can also reroute their alerts through other entry nodes and around congested edges, so the UAV and the path moves complement each other instead of mattering only for isolated sources.

Figure 7 shows that the search saturates quickly. SBS reaches 0.513 with $E = 250$ and 0.547 with $E = 1000$, and it rises further to the default $E = 2000$ but changes little beyond it, reaching 0.560, because the search stops after about 1897 evaluations on average, once a round accepts no further change. SBS-R follows the same trend, with 0.518, 0.546, and 0.562 for the same budgets. The default budget $E = 2000$ therefore lies in the saturated regime, and the lack of improvement of SBS-R over SBS cannot be attributed to an insufficient budget. Beyond the default budget,

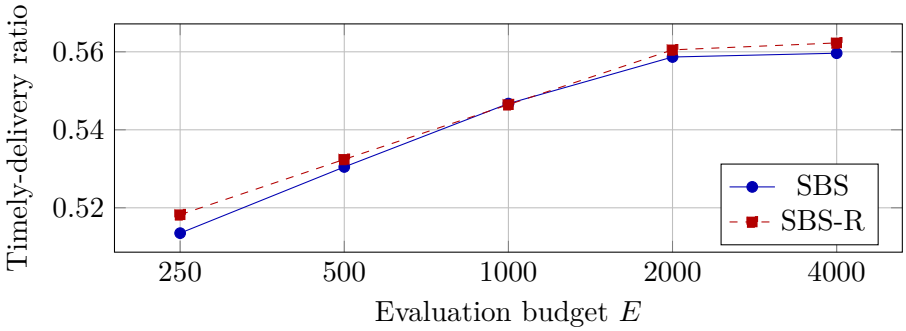



**Figure 7.** Timely-delivery ratio versus the evaluation budget on Wide.

further evaluations add almost nothing, whereas the number of design realizations still matters. Running SBS-R on the first replicate of each Wide topology, designing on expected rates yields 0.492, one realization 0.488, five realizations 0.529, and ten realizations 0.524, while the planning time grows from 34.1 to 262.7 seconds.

**Table 11.** Tightness of the LP bound on reduced instances.

| Scenario | $\|A\|$ | FixedTour | MILP plan | MILP | MILP bound | LP bound | Time (s) |
|---|---|---|---|---|---|---|---|
| Agis, replicate 3 | 10 | 9 | 9 | 9 | 9.00 | 9.51 | 3.1 |
| Belnet2008, replicate 1 | 10 | 5 | 5 | 5 | 5.00 | 6.41 | 2.1 |
| BtNorthAmerica, replicate 1 | 10 | 2 | 2 | 2 | 2.00 | 3.89 | 4.4 |
| Digex, replicate 2 | 10 | 1 | 1 | 1 | 1.00 | 2.38 | 1.1 |
| Digex, replicate 3 | 10 | 3 | 3 | 4 | 4.00 | 5.32 | 5.3 |

Table 11 compares the LP bound (P2) with the exact MILP (P3) on 5 reduced instances with ten alerts each, using the FixedTour trajectory and the first channel realization; HiGHS solved 5 of these MILPs to optimality. The plan built from the MILP solution is executed with the better of its own bandwidths and backlog-based equal sharing. On these instances, FixedTour attains the MILP optimum in 4 cases and the plan built from the MILP solution in 4 cases, so the gap between the LP bound and the achieved value comes mainly from relaxing the integrality of path selection. The bandwidth rule matters as well: taking the bandwidths directly from the MILP solution delivers only 4 timely alerts in total over the five instances, whereas keeping the solution paths and sharing bandwidth equally by backlog delivers 20, because a static bandwidth schedule falls out of step with how the EDF scheduler serves the queues. Finally, the envelope of ten tangents overestimates the rate by at most 0.0242 in relative terms on $[B_{\text{tot}}/512, B_{\text{tot}}]$. The gaps in Table 5 should therefore be read as upper bounds on the room for improvement, and the improvement that is actually achievable may be considerably smaller.

In terms of computational cost, NoUAV plans almost instantly, and FixedTour needs 0.210 seconds per Wide scenario on average because it simulates every candidate. SBS and SBS-R take 172.3 and 181.1 seconds with 1861 and 1975 evaluations on average, running 12 processes in parallel, and evaluating one plan on one realization takes about 0.043 seconds. The 3600 LPs that bound NoUAV and FixedTour have a median of 23203 and at most 74062 variables, and their solve time has a median of 1.85 seconds and a maximum of 44.4 seconds.

## 6.4. Sensitivity and RescueNet Case Study

Figure 8 varies one parameter at a time around the Wide setting on the first replicate of each evaluation topology; the middle point of each panel corresponds to Wide, except in the backhaul panel, where the right and left points are Wide and Narrow. FixedTour remains below SBS

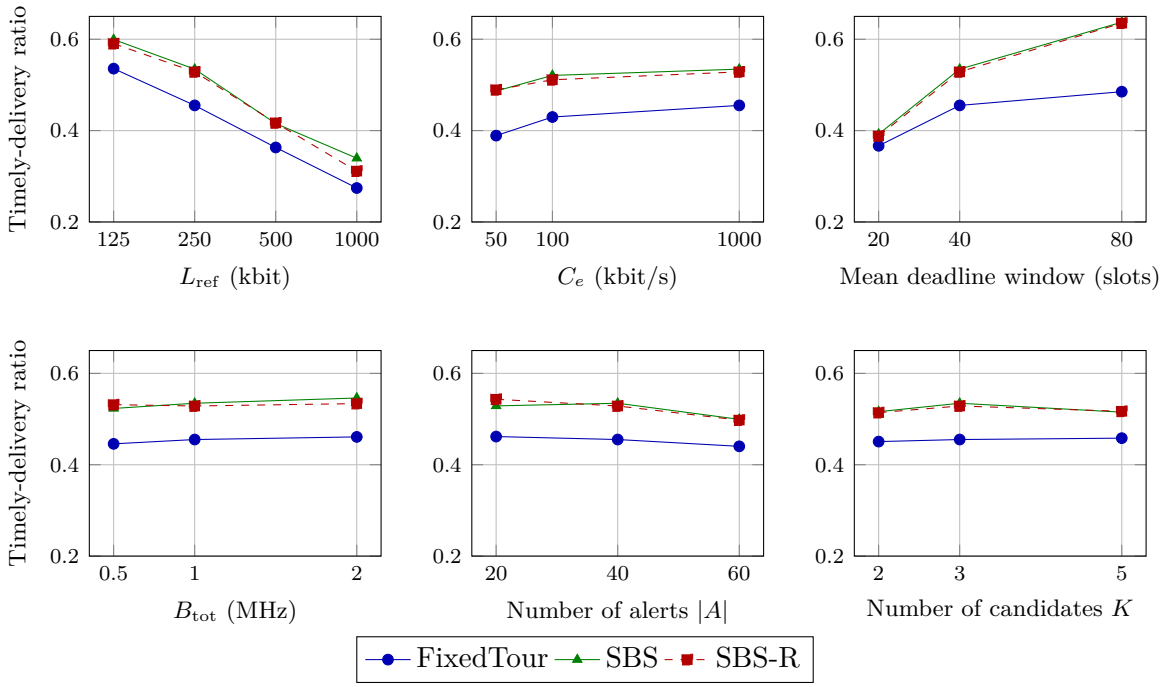



**Figure 8.** Sensitivity of the timely-delivery ratio to the main parameters.

and SBS-R in every non-degenerate family. The advantage narrows with short deadlines, where FixedTour reaches 0.367 and SBS-R 0.388, and widens with long deadlines, where FixedTour reaches 0.485 and SBS-R 0.635. Larger alerts reduce the ratio of every method sharply; with $L_{\text{ref}} = 1\,\text{Mbit}$, FixedTour reaches 0.274 and SBS-R only 0.311. Raising the number of alerts to 60 lowers the ratio of SBS-R only moderately, to 0.498 compared with 0.529 on Wide, and halving $B_{\text{tot}}$ barely changes it, at 0.532. The number of candidates $K$ has little influence on SBS and SBS-R, as expected: $K$ now only shapes the initial plan of FixedTour and the restarts of unserved alerts, since the path block generates further routes on its own. The family with stronger channels, which uses $\beta_0 = -40\,\text{dB}$ and $N_0 = -164\,\text{dBm/Hz}$, is degenerate: every source has a ground relay in range and all three methods reach 1.000, so we do not use it to draw conclusions about the role of the UAV.

**Table 12.** Results of the RescueNet case study.

| **Method** | **Timely ↑** | **95% CI** | **Conn ↑** | **Gap ↓** | **Planning (s) ↓** |
|---|---|---|---|---|---|
| NoUAV | 0.312 | [0.235, 0.393] | 0.376 | 0.142 | 0.0 |
| FixedTour | 0.395 | [0.313, 0.475] | 1.000 | 0.324 | 0.2 |
| SBS | 0.468 | [0.391, 0.537] | 1.000 | 0.258 | 225.0 |
| SBS-R | 0.471 | [0.397, 0.539] | 1.000 | 0.247 | 229.2 |
| Tran-IA | 0.450 | [0.381, 0.523] | 1.000 | 0.373 | 375.8 |

The RescueNet case study in Table 12 reproduces the ranking of the main experiment. Without a UAV, NoUAV delivers only 0.312 of the alerts on time, and only 0.376 of the sources have a time-respecting path to the center. The fixed trajectory of FixedTour raises the ratio to 0.395, and SBS, SBS-R, and Tran-IA reach 0.468, 0.471, and 0.450 with overlapping confidence intervals. As before, jointly optimizing trajectory and paths improves on the fixed trajectory and repair adds nothing, while Tran-IA comes close to SBS but with a larger gap, 0.373 instead of 0.258, and a longer planning time, 375.8 instead of 225.0 seconds. Although fewer sources have a

ground relay in range than in Wide, the gains of FixedTour and SBS over NoUAV remain close to those of the main experiment.

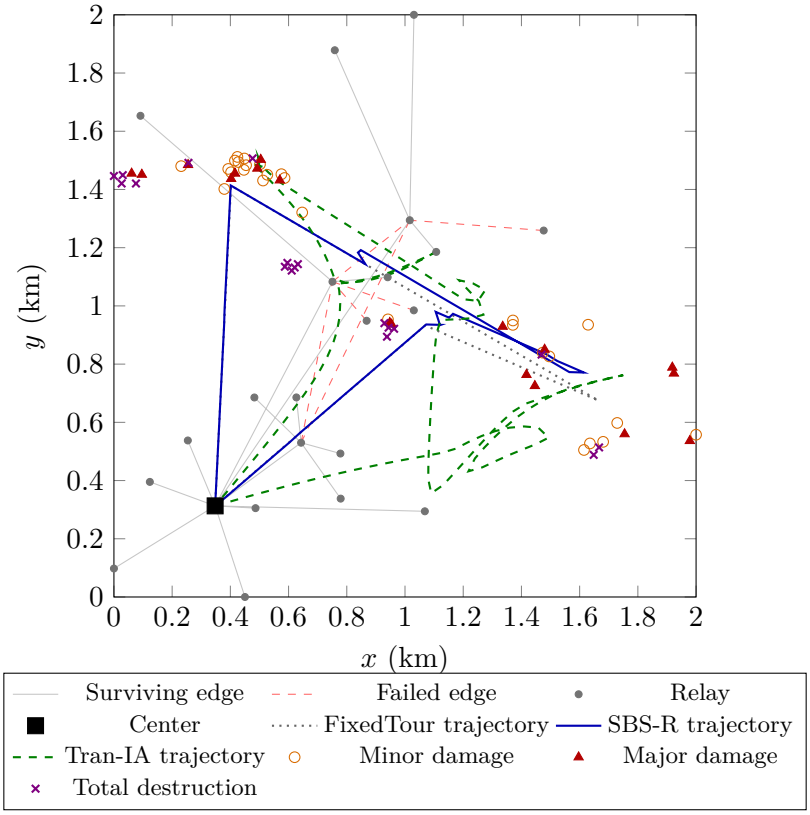



**Figure 9.** Targets, relay network, and UAV trajectories in the representative case-study scenario.

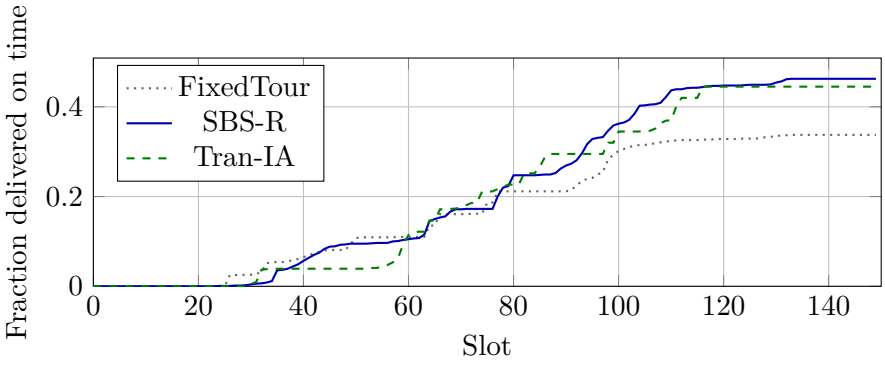



**Figure 10.** Delivery progress in the representative case-study scenario.

Figure 9 depicts scenario Garr200404, replicate 1, chosen because the timely-delivery ratio of SBS-R on it is closest to the median. SBS-R still flies a tour of the damage zones, but its path carries the small detours produced by the segment shifts, which pull the UAV toward target clusters whose uplinks are saturated, whereas Tran-IA follows a continuous path that reaches farther toward both sides of the target region. On this scenario, FixedTour, SBS-R, and Tran-IA deliver 0.338, 0.463, and 0.445 of the alerts on time. Figure 10 plots the fraction of the 40 alerts delivered on time by the end of each slot, averaged over 30 realizations, and shows that all three methods deliver alerts in bursts as the UAV approaches the source clusters; FixedTour almost stops after slot 100, whereas SBS-R and Tran-IA continue to deliver alerts with late deadlines and finish higher.

# 7. Conclusions

This paper studied the joint optimization of UAV trajectory, relay selection, and bandwidth allocation for alerts with delivery deadlines in a damaged ground network with finite backhaul. We proved that the problem is NP-hard even when the trajectory is fixed, derived an LP upper bound that is valid for every channel realization and every plan within the candidate set, and proposed a block search that scores candidate plans on sampled channel realizations. On 30 evaluation scenarios with regular backhaul, this method delivers 0.559 of the alerts on time, compared with 0.494 for a UAV on a fixed trajectory and 0.413 without a UAV, with differences of the same sign on all ten topologies, and the gain is larger under narrow backhaul. The block search also outperforms an adaptation of the method of Tran et al. [2], although the differences are not significant after Holm correction, and the ranking of the methods is preserved in a case study with targets extracted from RescueNet imagery.

The experiments also clarify which design choices matter. The local path moves are used for a substantial share of the alerts, and the local trajectory moves come close to a full scan of predefined tours while spending far fewer evaluations per round. Once the EDF scheduler shares bandwidth according to the backlog, the decisive decisions are which path each alert takes and where the UAV hovers, so explicit bandwidth weights and a dedicated risk-driven repair add no measurable gain; the repair operators act on the same variables that the block search has already refined. Designing on sampled channel realizations matters more than raising the search budget beyond its default, consistent with the observation of [4] that the rate at the average channel overestimates the expected rate, and the same effect explains why Tran-IA, which designs on a deterministic effective channel without knowledge of the backhaul, loses most of its advantage when the backhaul is narrow. Finally, the leximin objective of SBS-Lex lowers the timely-delivery ratio sharply, which shows that the max–min objective of the reference work is ill-suited to deadline-constrained alerts.

It is worth noting that the proposed method is a heuristic that only guarantees a non-decreasing objective during the search, not optimality. The relative gaps of SBS-R and SBS to the envelope bound are still 0.205 and 0.204, but most of this gap on small instances stems from the integrality relaxation of path selection, so a tighter bound is needed before judging how close the method is to optimal. The results also rely on simulation with independent channel states across slots, fixed powers, ten topologies, and a single case-study layout with synthetic alert times and sizes. Because the search may route alerts outside the candidate set, the LP bound measures the gap only relative to plans within that set; a bound over all routes would require a flow formulation on the whole relay graph. Future work includes such tighter bounds, a better mix of local and large trajectory moves, repair operators that reroute several alerts at once, design with more realizations within the same compute time, and case studies with measured victim locations or traffic.

## Data and Code Availability

The source code, scenario manifests, experiment configurations, and raw results are available in the project repository, together with scripts that regenerate every table and figure of this paper from the stored results. Topology Zoo, SNDlib, and RescueNet are publicly available from their original sources; the RescueNet images and masks are not redistributed, and only the extracted target manifest is provided.

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